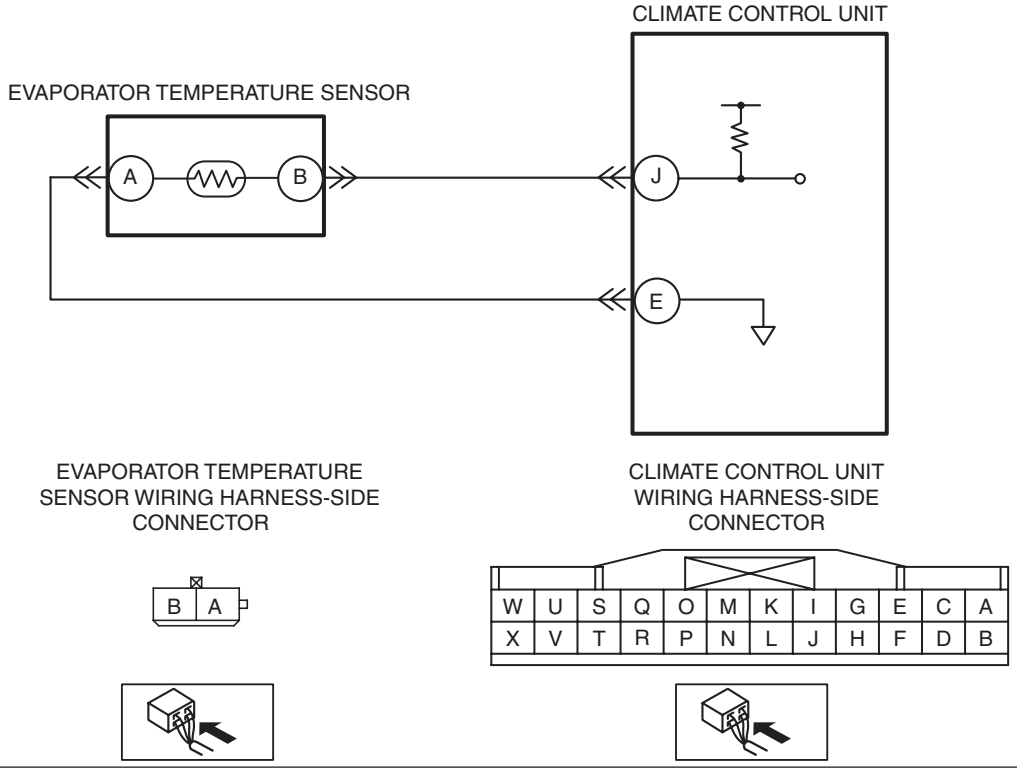
DTC B1B71:11, B1B71:13 [FULL-AUTO AIR CONDITIONER]

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System malfunction location	• B1B71:11: Evaporator temperature sensor circuit short to ground • B1B71:13: Evaporator temperature sensor circuit open
Detection condition	• Malfunction in wiring harness between evaporator temperature sensor and climate control unit
Fail-safe function	• A/C output OFF is controlled when the evaporator temperature sensor input value is at **0 °C {32 °F}**.
Possible cause	• Connector or terminal malfunction • Evaporator temperature sensor malfunction • Open circuit in wiring harness between climate control unit and evaporator temperature sensor • Short to ground in wiring harness between climate control unit and evaporator temperature sensor • Climate control unit malfunction



Diagnostic procedure

STEP	INSPECTION		ACTION
1	**INSPECT EVAPORATOR TEMPERATURE SENSOR CONNECTOR** • Switch the ignition off. • Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Disconnect the evaporator temperature sensor connector. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Inspect the connector and terminals (corrosion, damage, pin disconnection). • Are the connector and terminals normal?	Yes	Go to the next step.
		No	Repair/replace the connector or terminal. After repair procedure, go to the next step.
2	**INSPECT EVAPORATOR TEMPERATURE SENSOR** • Inspect the evaporator temperature sensor. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Is it normal?	Yes	Go to the next step.
		No	Replace the evaporator temperature sensor. (See A/C UNIT DISASSEMBLY/ASSEMBLY.) Go to the next step.

STEP	INSPECTION		ACTION
3	**INSPECT EVAPORATOR TEMPERATURE SENSOR CIRCUIT FOR OPEN CIRCUIT** • Disconnect the climate control unit connector. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) • Disconnect the evaporator temperature sensor connector. (See EVAPORATOR TEMPERATURE SENSOR INSPECTION [FULL-AUTO AIR CONDITIONER].) • Inspect for continuity between the following terminals (wiring harness-side): — Climate control unit terminal E—evaporator temperature sensor terminal A — Climate control unit terminal J—evaporator temperature sensor terminal B • Is there continuity?	Yes	Go to the next step.
		No	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and evaporator temperature sensor terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for an open circuit. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has an open circuit. Go to the next step.
4	**INSPECT EVAPORATOR TEMPERATURE SENSOR CIRCUIT FOR SHORT TO GROUND** • Inspect for continuity between the following terminal (wiring harness-side) and body ground: — Climate control unit terminal J • Is there continuity?	Yes	Refer to the wiring diagram and verify whether or not there is a common connector between climate control unit terminal and evaporator temperature sensor terminal. **If there is a common connector:** • Determine the malfunctioning part by inspecting the common connector and the terminal for corrosion, damage, or pin disconnection, and the common wiring harness for a short to ground. • Repair or replace the malfunctioning part. **If there is no common connector:** • Repair or replace the wiring harness which has a short to ground. Go to the next step.
		No	Connect the climate control unit connector, then go to the next step.
5	**VERIFY CLIMATE CONTROL UNIT CONNECTOR CONDITION** • Inspect the connector and terminals (corrosion, damage, pin disconnection). • Are the connector and terminals normal?	Yes	Go to the next step.
		No	Repair/replace the malfunctioning vehicle wiring harness, connector, or terminal. After repair procedure, go to the next step.
6	**INSPECT EVAPORATOR TEMPERATURE SENSOR CIRCUIT** • Connect the climate control unit connector. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Switch the ignition ON (engine off or on). • Inspect the voltage at the following climate control unit terminal (wiring harness-side). — Terminal J (evaporator temperature sensor input signal) • Is the voltage normal? (**Approx. 5 V**)	Yes	The system is normal at present. Go to the next step.
		No	Go to the next step.
7	**VERIFY THAT SAME DTC IS NOT OUTPUT AGAIN** • Switch the ignition off. • Disconnect the negative battery cable. • Reconnect the disconnected connectors. • Connect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.) • Clear the past malfunction from memory. (See CLEARING DTC [FULL-AUTO AIR CONDITIONER].) • Verify DTCs. (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Is DTC B1B71:11 or B1B71:13 output?	Yes	Repeat the inspection from Step 1. • If the malfunction does not recur, go to the next step. • If the malfunction recurs, replace the climate control unit. (See CLIMATE CONTROL UNIT REMOVAL/ INSTALLATION [FULL-AUTO AIR CONDITIONER].) Go to the next step.
		No	Go to the next step.

STEP	INSPECTION		ACTION
8	**VERIFY THAT NO OTHER DTCs ARE PRESENT** • Verify other DTCs displayed. (See DTC DISPLAY [FULL-AUTO AIR CONDITIONER].) • Are any other DTCs output?	Yes	Perform the corresponding DTC inspection. (See DTC TABLE [FULL-AUTO AIR CONDITIONER].)
		No	DTC troubleshooting completed.